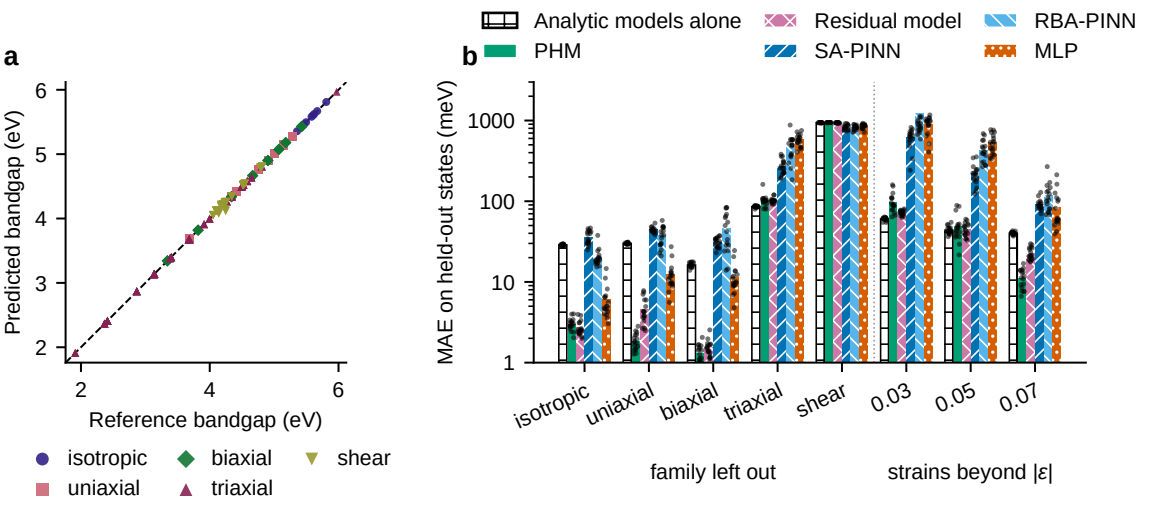

a
Predicted bandgap (eV)
Reference bandgap (eV)
isotropic
biaxial
shear
uniaxial
triaxial
b
Analytic models alone
PHM
Residual model
SA-PINN
RBA-PINN
MLP
MAE on held-out states (meV)
isotropic
uniaxial
biaxial
triaxial
shear
0.03
0.05
0.07
family left out
strains beyond |ε|